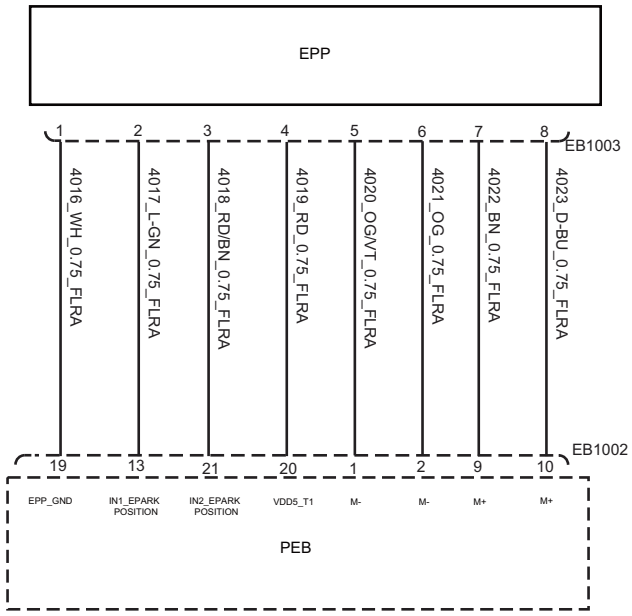
P1A7C-19、 P1A44-19

Fault diagnosis code
P1A7C-19：EPP overcurrent fault
P1A44-19：EPP mos tube overcurrent fault
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • EPP module fault • The EPP controller circuit in the PEU is faulty
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1A7C-19
Absolute current over 18A for 10ms or more than 15A lasts for I^2*0.04Ω*t=1J (open tube entry diagnostics)
P1A44-19
To set the effect of a fault code condition
Parking system malfunction

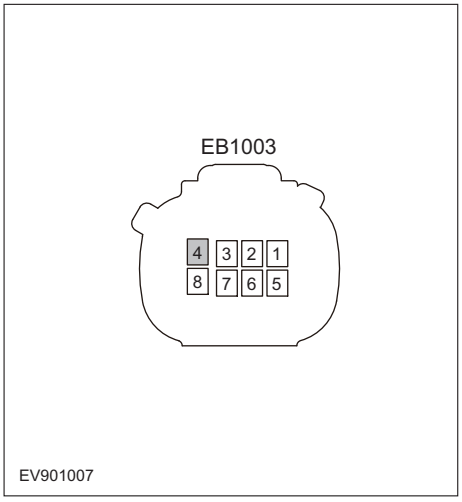
Circuit diagram

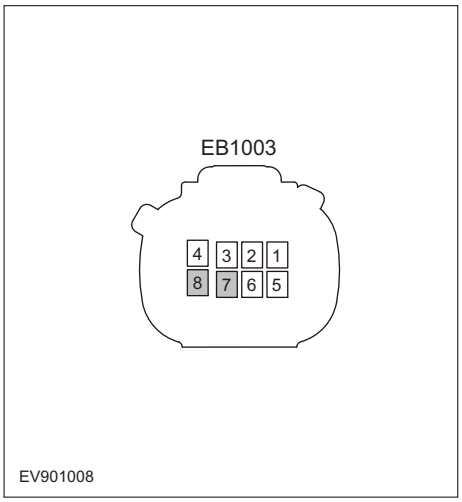


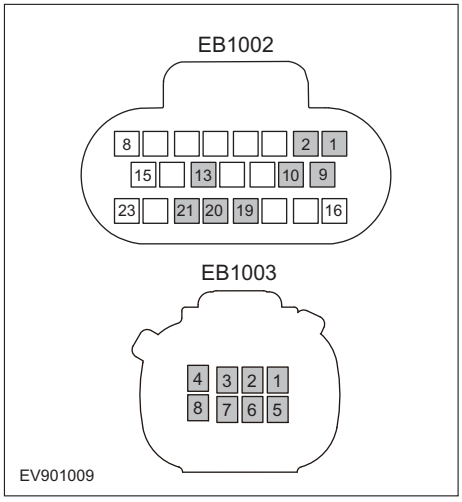


EV901006

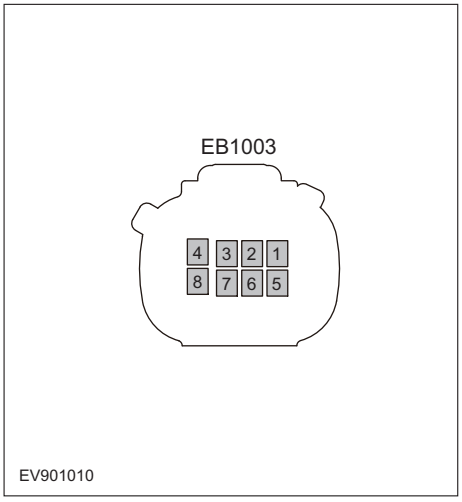
Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of PEU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the PEU with diagnostic apparatus. 4 If there is any fault code except for P1A7C-19,P1A44-19?
		YES → Refer to:DTC Summary list(PEU).
NO ↓		
Step 3	Perform initialization.	
		1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The governor was initialized EPP with the diagnostic instrument. 5 EPP can the governor be initialized and is the governor restored?
		NO → Repair or change the EPP governor.
YES ↓		

Step 4	Check the power supply circuit of EPP circuit.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1003 of EPP. 3 Turn the ignition switch to ON. 4 Measure the voltage between the harness terminal of EPP and ground. Measuring circuit: voltage between terminal 4 on plug EB1003 and ground. **Standard value:4.8~5.2V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the supply power of EPP when it is started or operating.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1003 of EPP. 3 Turn the ignition switch to ON. 4 Turn on the EPP control switch to make the EPP motor in working condition. 5 Measure the voltage between the harness terminal of EPP and ground. Measuring circuit: voltage between terminal 7 on plug EB1003 and ground. Measuring circuit: voltage between terminal 8 on plug EB1003 and ground. **Standard value:10~14V** 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the EPP circuit(open).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB1002 of PEU module. 3 Measure the resistance between the harness terminal of EPP and the PEU. Measuring circuit: resistance between terminal 1 on plug EB1003 and terminal 19 on plug EB1002. Measuring circuit: resistance between terminal 2 on plug EB1003 and terminal 13 on plug EB1002. Measuring circuit: resistance between terminal 3 on plug EB1003 and terminal 21 on plug EB1002. Measuring circuit: resistance between terminal 4 on plug EB1003 and terminal 20 on plug EB1002. Measuring circuit: resistance between terminal 5 on plug EB1003 and terminal 1 on plug EB1002. Measuring circuit: resistance between terminal 6 on plug EB1003 and terminal 2 on plug EB1002. Measuring circuit: resistance between terminal 7 on plug EB1003 and terminal 9 on plug EB1002. Measuring circuit: resistance between terminal 8 on plug EB1003 and terminal 10 on plug EB1002. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 7	Check the circuit of EPP (short).



1 Turn the ignition switch to OFF.

2 Measure the resistance between the harness terminal of EPP and ground.

Measuring circuit: resistance between terminal 1 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 2 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 3 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 4 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 5 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 6 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 7 on plug EB1002 and ground.

Measuring circuit: resistance between terminal 8 on plug EB1002 and ground.

Standard value: ∞

3 Measure the voltage between the harness terminal of EPP and ground.

Measuring circuit: voltage between terminal 1 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 2 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 3 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 4 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 5 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 6 on plug EB1002 and ground.

Measuring circuit: voltage between terminal 7 on plug EB1002 and ground.

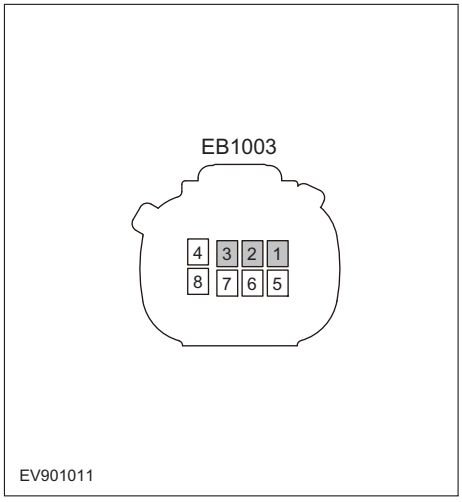
Measuring circuit: voltage between terminal 8 on plug EB1002 and ground.

Standard value: ≈ 0V

4 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 8	Check the EPP (components).
	1 Measure the resistance of the EPP terminal directly. Measuring circuit: resistance between terminal 2 and terminal 1 on plug EB1003. Measuring circuit: the resistance between terminal 3 and 1 on plug EB1003. **Standard value:** 2 Check whether the result is normal or not?
	NO → Replace the EPP.
YES ↓	
Step 9	Check EPP controller circuit.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check EPP controller circuit. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 10	Test whether PEU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1002 of PEU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
	YES → Change the PEU Module
NO ↓	

At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.
Next Step ↓
Diagnosis end.